

The outcome of the UAV Formative Assessment

- Understanding the elements of a Drone.
- Understanding the history of Unmanned Aerial Vehicle (UAV).
- Understanding the Drone Revolution.
- Understanding UAV's can resolve challenges.
- Understanding the home application of drones as an individual.
- Understanding the commercial applications of drones as an entrepreneur.
- Understanding the Different Careers and businesses as a licensed Drone Pilot.
- Understanding the Laws and Policy of UAV in South Africa.
- Understanding the Development of Drones.

Section A – The Drone Diagram

In the Diagram below place the correct labels to the numbers that are given in the Diagram.

For example 1. GPS ETC. (21)

The Following words for the Aircraft Diagram are as follow:

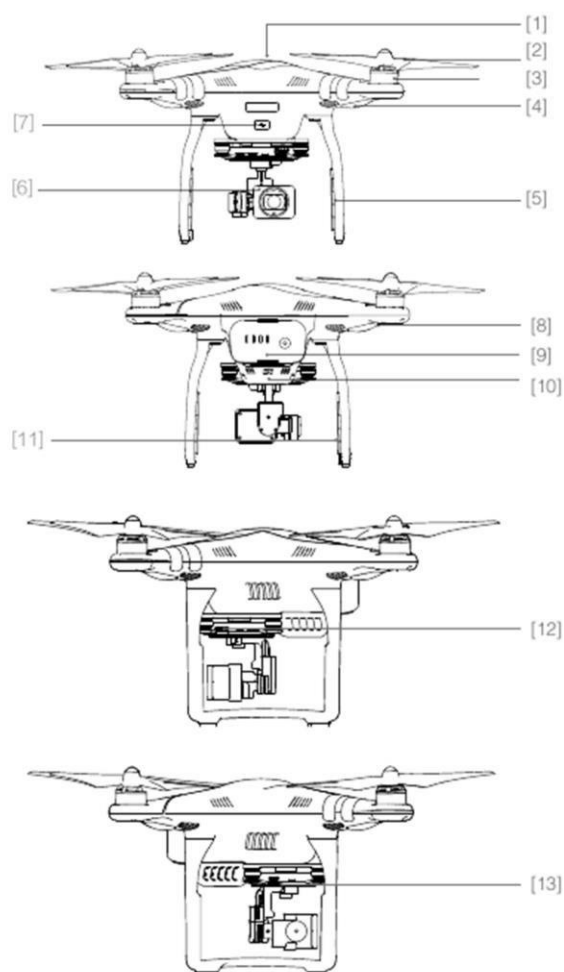
Propeller, Front LED, Motor, Gimbal and Camera, Landing Gear, Aircraft status indicator, Aircraft Micro – USB port, Visioning positioning sensor, Intelligent Flight Battery, Camera Micro-Card SD Slot, Antennas, Camera Micro-USB port

The Following words for the Remote Controller Diagram are as follow:

Return Home Button (RTH), Antennas, Control stick, Mobile Device Holder, Status LED, Battery Level LEDs, RTH LED, Power Button.

Use the words given above to complete the Aircraft diagram and Remote Controller Diagram.

Aircraft Diagram



1. GPS

2.

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4.

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7.

8.

9.

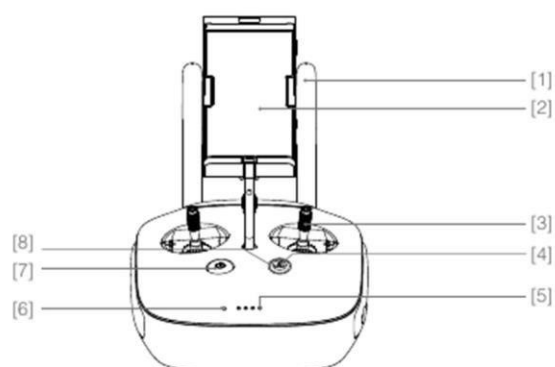
10.

11.

12.

13.

Remote Controller Diagram



1.

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Section B – Challenges that will be solved by UAV (40)

1. List the challenges that you face in your community that a drone may be able to solve and clarify why? (8)
2. List the solutions to the challenges you have identified? (8)
3. Your understanding of how you see Autonomous Vehicles being rolled out in your community, specifying if it will be consumer or industrial Autonomous Vehicles.? (8)
4. List the UAV services that will be needed in your community?(8)
5. What impact does the UAV have on the Humanitarian and Disaster Relief? (6)

Section C - The Different Levels and Applications of UAV (30)

1. Name 4 Home use applications of UAV and specify what use does a UAV do? (10)
2. Name 6 Commercial Applications that will benefit your community and state why would they benefit your community? (12)
3. What is the difference between Home use application and Commercial application? (8)

Section D – The Laws and Policy of UAV in South Africa (30)

1. What does RPL stand for? (2)
2. Which Authority oversees the laws of the UAVS? (2)
3. What does remotely piloted aircraft (RPAS) mean in your own words? (2)
4. Which aircraft is a product falling under the definition of aircraft which is designed or intended for use in play by children? (2)
5. What is a Model aircraft mean? (2)
6. In regards to private use a remotely piloted aircraft may only be used for what? (4)
7. In regards to all other uses a RPAs may only be used for what? (4)
8. Name two dangers of negligent operations of RPAS? (4)
9. List the Do's and Don'ts of a UAV? (8)

Section D – Business Plan (30)

Description

Prepare a business plan detailing the following:

- List the Autonomous Vehicles services that you think are needed in the community.
- Your ideal pricing structure for each of the service you plan to provide.
- Major Internet challenges you see delivering your business plan.
- Solutions to the challenges you have identified.
- List the stakeholders you think will be needed to resolve the challenges.

Section E – Identifying The Needs Of Drones Within Your Community (30)

Description

Prepare a detailed document on what drones can do to assist your community.

- Identifying the problem.
- How will the drone assist with the identified problem.
- Describe the planning around the implementation of the drone as a solution.
- The Ideal pricing.